MGT 207
Production and Operations Management

Introduction

Operations vs. Other Business Functions

- Marketing
  - ?
- Finance
  - ?
- Operations Management

3 Flavors of Operations Mgmt

- Operations Research (aka Business Analytics)
  - Optimization
  - Network design
  - Vehicle routing
  - Game theory
- Operations Management
  - Operations management takes tools from Operations Research and applies them to business settings
- Data Analytics
  - Assortment design
  - Revenue management

Operations Research: An Example

You want your diet to meet some nutrition standards. According to your daily diet plan you need to have

at least 100 percent of the U.S. RDA of vitamin C, calcium; at least 55 grams of protein; and at most 2000 calories.

You are wondering if this can be accomplished by eating at McDonald's. Can you design a least-cost McDonald's daily meal plan that meets your daily nutritious standard?

McDonald's Food: the Facts

Menu Item	Price* ($)	Calories	Protein	Fat	Sodium	Vit. A	Vit. C	Calcium	Iron
			(g)		(mg)	% U.S. RDA			
Hamburger	0.69	**260**	**13**	9	530	2	**2**	**15**	15
Big Mac	1.99	**560**	**25**	30	1010	8	**2**	**25**	25
Filet-O-Fish	1.89	**400**	**14**	18	640	2	**0**	**15**	10
Small French Fries	1.39	**250**	**2**	13	140	0	**6**	**2**	4
Chicken McNuggets (6pcs)	1.99	**250**	**15**	15	670	2	**2**	**2**	4
Garden Salad	2.05	**35**	**2**	0	20	120	**40**	**4**	6
Honey	0	**45**	**0**	0	0	0	**0**	**0**	0
Baked Apple Pie	0.79	**260**	**3**	13	200	0	**40**	**2**	6

* Prices recorded Jan. 12, 2000 in St. Louis, Missouri

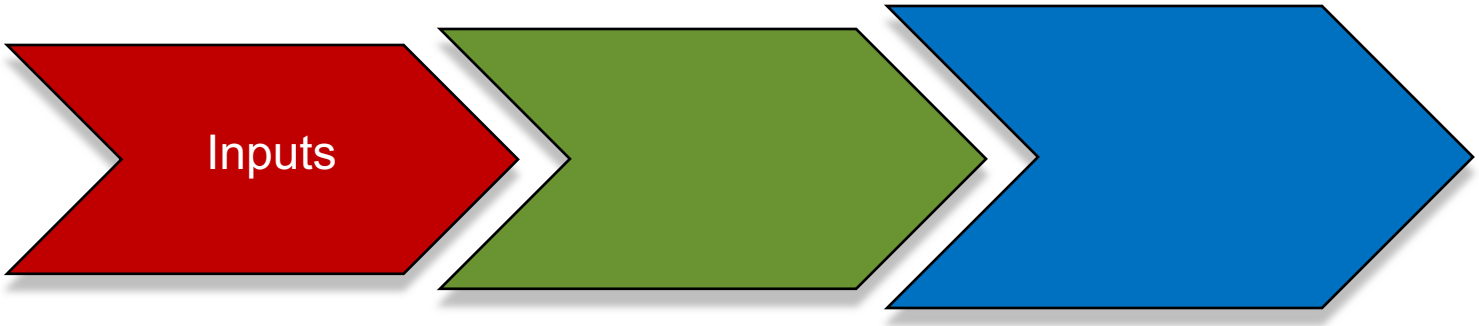
Operations Management: Definition and Examples

A set of activities dedicated to the transformation of inputs into outputs of greater value.

- Core of most business organizations
- Responsible for the creation of an organization's goods or services.
- Involves managing transformations.
- Converts inputs into outputs using one or more transformation processes.
- Its essence is to *add value* during the transformation process.

Operations Management

A model of any kind of operation



Resources

- People
- Materials
- Equipment
- Knowledge
- Infrastructure

Operations Management

A model of any kind of operation

Resources

- People
- Materials
- Equipment
- Knowledge
- Infrastructure

Operations

- Assembly
- Transportation
- Extraction
- Cultivation
- Fabrication

Operations Management

A model of any kind of operation

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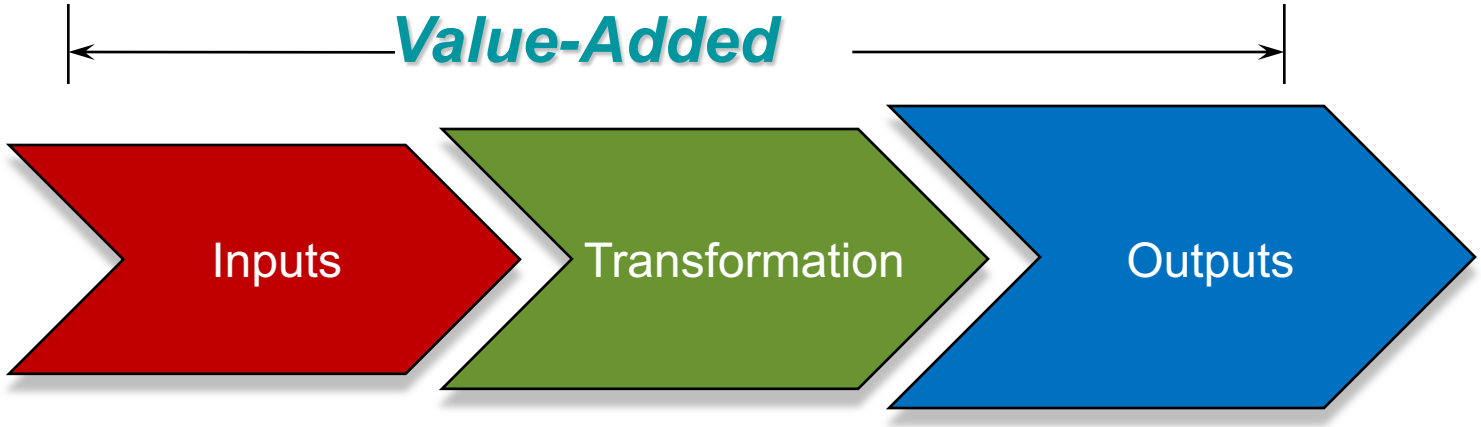
Outputs

- Goods and Services
- Pollution

Value-Added

The difference between the total value of the outputs and the total value of the inputs associated with an operation.



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Types of Transformation

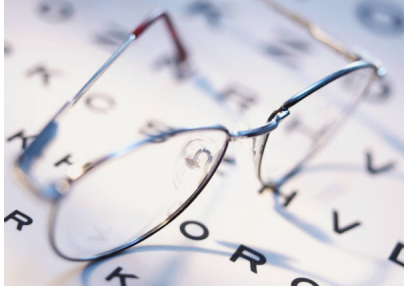
- Physical-manufacturing
- Locational-transportation
- Exchange-retailing
- Storage-inventory
- Physiological-health care
- Informational-telecommunications

Classifying an Operation

- *Goods:* Tangible products-Manufacturing Operations
- *Services:* Intangible products-Service Operations

Issues that Arise in Tangible Operations

- Inventory
- Factory Capacity
- Production Leadtime
- Quality Management



Operations of Goods Example

- At his ice cream stand, Art sells ice cream. He cannot foretell the number D (assume uniform(30,150)) of servings he can sell in any given day. Before Art's customers appear, however, he must produce his ice cream at a cost of $1.50 per serving. For each serving Art sells, he receives $4.50, its retail value. What Art does not sell must be thrown away at the end of the day. Art wants to maximize his average profit. How many servings should he produce every day?

Service Operations Example

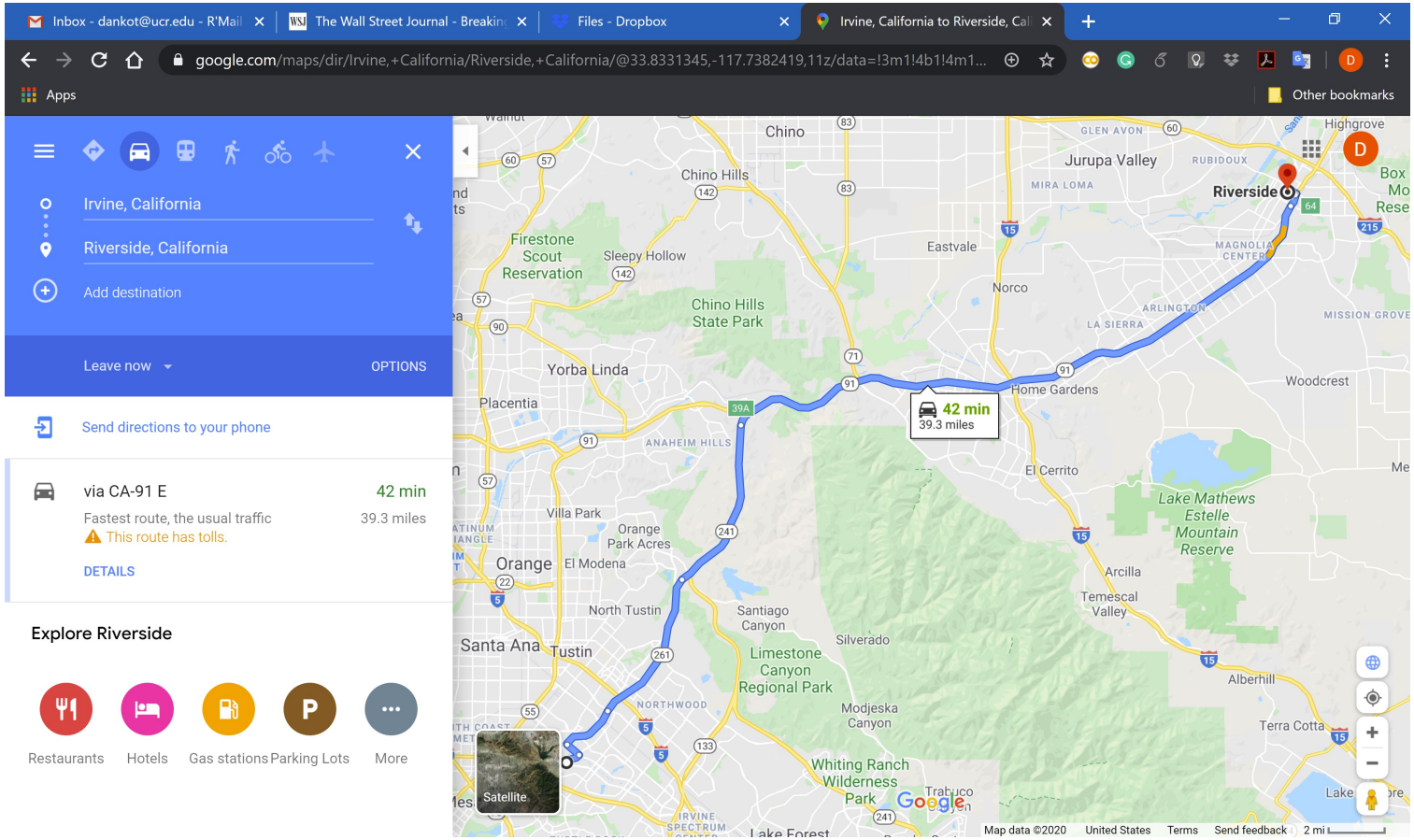
- A small railway station has a single ticket counter. During the peak hours, customers arrive at the rate of 10 per hour. The average number of customers that can be served is 12 per hour. Find out the following:
  - Probability that the ticket counter is free.
  - Average number of customers in the queue.

Other OM Applications (Integer Optimization)

UBER

amazon.com
Prime

Other OM Applications (Networks)



Google uses ILP to determine driving directions

Ever Notice That UPS Trucks Rarely Make Left Turns?

- An estimated **90%** of the turns made by UPS delivery trucks are right turns, and that's intentional, according to the *Washington Post*. Left turns are seen as inefficient, because they leave trucks sitting in traffic longer. The logistics company says a policy of minimizing left turns has helped it save more than **10 million** gallons of fuel over the past decade. Left turns (in countries where people drive on the right) are dangerous, too: New York City officials say left turns are **3 times more likely** than right turns to cause a deadly crash involving a pedestrian.

Course Objectives: The Student Should….

- provide you with an understanding of the operations and the role they play within an organization
- provide you with an understanding of how operations can provide a competitive advantage in the marketplace
- provide you with a knowledge of the issues related to designing and managing operations and the techniques to do so
- provide you the ability to use analytical tools and conceptual frameworks to guide your thinking about operations